

Assay-labelled evidence

Legacy DNA/lentiviral MPRA
Direct-RNA IRES-TrAPPr
In-cell structure atlas
IRES--cargo measurements

Task 0: audit

Native prediction benchmark
Cluster/source holdouts
Cross-assay calibration
Applicability distance

Task 1: optimize

Full-length viral seeds
Function lower confidence bound
Ensemble preservation
Cargo crosstalk + diversity

Independent evaluation

Assay transfer
Structural robustness
Worst-case cargo context
Validity, novelty, efficiency

Optimization scorer is never the sole final evaluator; all comparisons use matched budgets.